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Original Article

Early positive drain cultures predict major morbidity independent of postoperative pancreatic fistula formation after pancreatoduodenectomy

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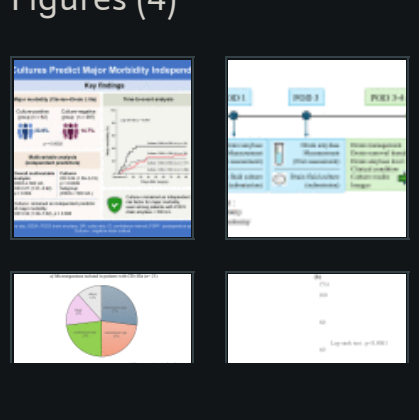


Abstract

Background

Early postoperative drain cultures are associated with intra-abdominal infectious complications after pancreatoduodenectomy (PD), but their value for predicting overall major morbidity and complementing conventional postoperative pancreatic fistula (POPF)-based risk stratification remains unclear. Therefore, this study evaluated the association between early postoperative drain culture positivity (Culture+) and major morbidity and assessed its utility for postoperative risk stratification following PD.

Methods

We retrospectively reviewed 449 consecutive patients who underwent PD with surveillance drain cultures obtained within postoperative day (POD) 3 between 2016 and 2025. The primary endpoint was major morbidity (Clavien–Dindo grade IIIa or higher). Independent predictors were identified using multivariable logistic regression, followed by a subgroup analysis in patients with POD3 drain amylase (D3DA) levels <500 U/L.

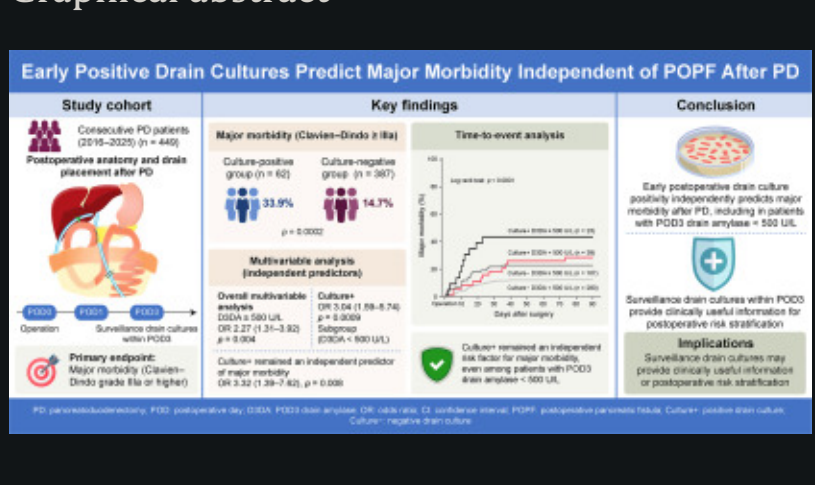
Results

Sixty-two patients (13.8%) were categorized as Culture+. Major morbidity occurred significantly more frequently in patients with Culture + than in those with Culture– (33.9% vs. 14.7%, p = 0.0002). Multivariable analysis identified D3DA ≥500 U/L (odds ratio [OR] 2.27, 95% confidence interval [CI] 1.31–3.92, p = 0.004) and Culture+ (OR 3.04, 95%CI 1.59–5.74, p = 0.0009) as independently associated with major morbidity. Among patients with D3DA <500 U/L, Culture + remained significantly associated with increased risk of major morbidity.

Conclusion

Early postoperative drain culture positivity is an independent predictor of major morbidity after PD. Patients with positive drain cultures remain at increased risk of major morbidity even when drain amylase levels are low, suggesting that surveillance drain cultures may provide clinically useful information for postoperative risk stratification.

Graphical abstract


Early Positive Drain Cultures Predict Major Morbidity Independent of POPF After PD
Study cohort
Key findings
Conclusion
Time-to-event analysis
Multivariable analysis (independent predictors)
Implications

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Keywords

Pancreatoduodenectomy; Early drain culture; Major morbidity; Clavien–Dindo classification; Infectious complication

1. Introduction

Despite advances in perioperative management and surgical techniques, pancreatoduodenectomy (PD) remains a highly invasive procedure and is associated with substantial postoperative morbidity. [1], [2], [3], [4] Major complications, defined as Clavien–Dindo (CD) grade IIIa or higher may also delay the initiation of adjuvant chemotherapy, ultimately affecting long-term oncologic outcomes.[3], [5] Accordingly, accurate early identification and appropriate surveillance of patients at high risk for major morbidity after PD remain important clinical challenges.

Among PD-related complications, postoperative pancreatic fistula (POPF) is widely regarded as the most clinically significant event, and contemporary postoperative management strategy focuses on POPF risk stratification.[6], [7] Early postoperative drain amylase levels (D-AMY) are widely used as a predictor of clinically relevant POPF, with several studies demonstrating the utility of D-AMY-guided early drain removal protocols.[8], [9], [10] However, D-AMY primarily reflects pancreatic leakage and may not fully capture the risk of major infectious complications unrelated to clinically relevant POPF.

In clinical practice, major infectious complications such as delayed intra-abdominal abscesses, infected fluid collections requiring drainage,[11] and post-pancreatectomy hemorrhage (PPH) may occur despite low D-AMY levels or in the absence of clinically relevant POPF.[12] Because PD involves biliary and gastrointestinal reconstruction, bacterial contamination of the operative field from the biliary tract or intestinal lumen may occur early after surgery and contribute to postoperative infectious complications.[13], [14]

Previous studies have investigated the relationship between preoperative bile cultures, intraoperative peritoneal cultures, and postoperative infectious complications.[15] Several reports have also suggested an association between postoperative drain cultures and intra-abdominal infections.[16] However, whether early postoperative drain cultures can predict overall major morbidity, particularly in relation to the timing of events, has not been fully elucidated. Furthermore, the clinical significance of drain culture results in stratifying infectious risk that cannot be captured by conventional POPF-centered assessments remains unclear.

Therefore, this study evaluated the association between early postoperative drain culture positivity and major morbidity following PD and investigated whether surveillance drain cultures provide complementary information for postoperative risk stratification beyond conventional drain amylase-based assessment.

2. Methods

2.1. Study design and patient population

This retrospective cohort study analyzed information from a prospectively maintained database at the study hospital. The study protocol was approved by the Ethics Committee of the study institution (approval number: R1721) and was performed in accordance with the Declaration of Helsinki. Informed consent was obtained using an opt-out approach. Inclusion criteria were all consecutive patients who underwent PD at our institution between January 2016 and December 2025 (n = 460). Patients without available postoperative surveillance drain fluid culture within postoperative day (POD) 3 were excluded (n = 11), resulting in a final analytic cohort of 449 patients. Clinical and perioperative data were retrospectively reviewed, and follow-up information was updated in May 2026.

2.2. Surgical technique and postoperative management

All PDs were performed under general anesthesia. At our institution, PD was indicated for all patients with resectable periampullary tumors when curative resection was considered feasible. Standard resection included the pancreatic head, duodenum, distal bile duct, and gallbladder. Surgeons performed portal vein or superior mesenteric vein resection and reconstruction when required for oncologic clearance. Reconstruction consisted primarily of duct-to-mucosa pancreatojejunostomy, followed by biliary and gastrointestinal reconstruction.[17]

At surgery completion, surgeons routinely placed closed suction drains around the pancreatojejunostomy and hepaticojejunostomy sites (Fig. 1a). Postoperative management followed standardized institutional protocols throughout the study period. Perioperative antimicrobial prophylaxis consisted of flomoxef sodium (FMOX), an oxacephem-class β-lactam antibiotic commonly used for surgical prophylaxis in Japan, immediately before skin incision, every 3 h intraoperatively, and until POD 2.

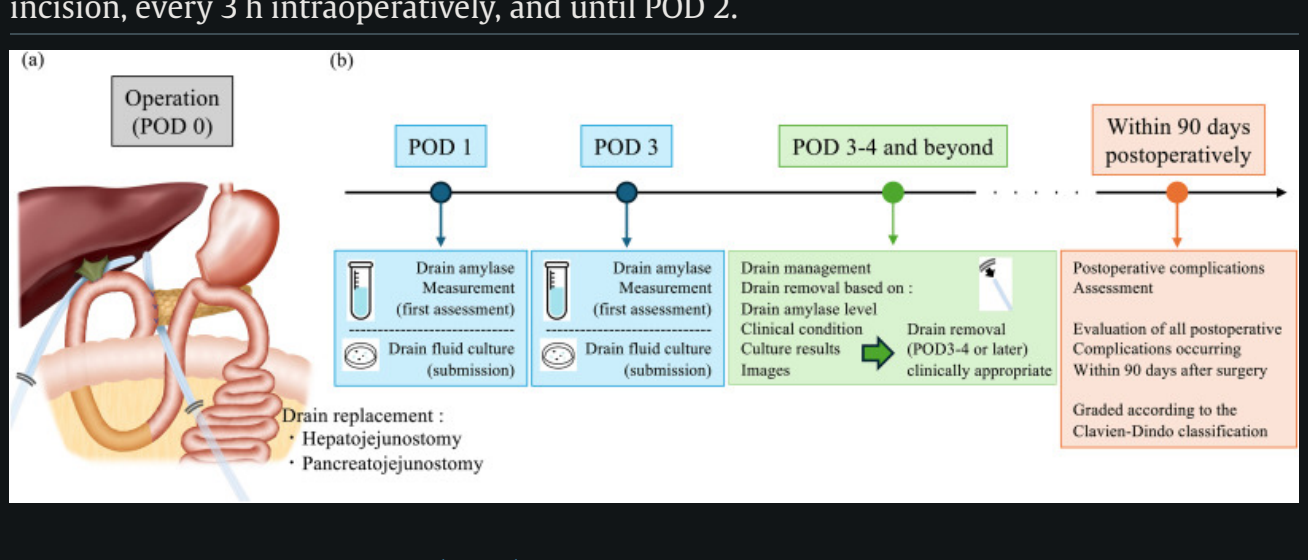
(a)
Operation (POD 0)
Drain replacement :
・Hepatojejunostomy
・Pancreatojejunostomy
(b)
POD 1
POD 3
POD 3-4 and beyond
Within 90 days postoperatively
Drain amylase Measurement (first assessment)
Drain fluid culture (submission)
Drain amylase Measurement (first assessment)
Drain fluid culture (submission)
Drain management
Drain removal based on :
Drain amylase level
Clinical condition
Culture results
Images
Drain removal (POD3-4 or later) clinically appropriate
Postoperative complications Assessment
Evaluation of all postoperative Complications occurring Within 90 days after surgery
Graded according to the Clavien-Dindo classification

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Fig. 1. **Drain placement and postoperative surveillance protocol after pancreaticoduodenectomy.**
(a) Schematic illustration of drain placement after reconstruction following pancreaticoduodenectomy. Closed-suction drains were routinely positioned around the pancreatojejunostomy and hepaticojejunostomy sites.

(b) Schematic overview of the postoperative surveillance protocol used in the present study. Drain amylase measurement and drain fluid culture were routinely performed on postoperative days (POD) 1 and 3. Drain removal was considered on POD3 or later according to drain findings and the patient's postoperative clinical condition. Postoperative complications were assessed within 90 days after surgery and graded according to the Clavien–Dindo classification.

The clinical team routinely measured drain fluid amylase levels after surgery and submitted surveillance drain fluid cultures on POD 1 and 3 to assess early intra-abdominal bacterial colonization (Fig. 1b). Clinicians performed additional microbiological examinations in patients who developed fever, elevated inflammatory marker levels, or clinical signs of intra-abdominal infection. The clinical team determined the timing of drain removal based on the overall clinical course, drain output characteristics, and laboratory findings, including drain amylase levels and culture results. Patients also underwent routine postoperative surveillance, including physical examinations, laboratory testing, and imaging studies as clinically indicated. Although our institutional policy favors early drain removal, positive culture alone did not require continued drainage. Drain removal was individualized based on clinical condition, drain output, and laboratory findings, particularly when output was low and serous without evidence of ongoing infection.

2.3. Data collection and definitions

We collected demographic characteristics, comorbidities, perioperative laboratory findings, operative variables, postoperative drain management data, microbiological culture results, pathological findings, and postoperative outcomes from electronic medical records.

Overall postoperative morbidity was graded according to the CD classification system. When multiple complications occurred in the same patient, the highest CD grade was used for analysis. Clinically relevant POPF (CR-POPF), delayed gastric emptying (DGE), and PPH were defined according to the criteria of the International Study Group of Pancreatic Surgery criteria.[7] Postoperative bile leakage was evaluated using the definition proposed by the International Study Group of Liver Surgery.[18] Early drain culture positivity (Culture+) was defined as the isolation of any microorganism, irrespective of species or polymicrobial growth, from drain fluid cultures obtained on or before POD3.

2.4. Study endpoints and definitions

The primary endpoint was the occurrence of major morbidity (CD grade IIIa or higher) within 90 days after surgery. All CD grade IIIa or higher events were reviewed using discharge summaries and outpatient records through postoperative day 90. Non-surgical events, such as oncologic disease progression, were excluded from the major morbidity endpoint. Secondary endpoints included the incidence of infectious complications, the incidence of CR-POPF, the time to onset of infectious complications, and the spectrum of microorganisms isolated from postoperative drain cultures.

2.5. Statistical analysis

Continuous variables are expressed as medians with interquartile ranges (IQRs), and categorical variables are summarized as frequencies and percentages. Continuous variables and categorical variables were compared using the Kruskal–Wallis test and Pearson's chi-square tests, respectively.

POD3 drain amylase (D3DA) was analyzed using receiver operating characteristic (ROC) curve analysis to identify the optimal cutoff value for predicting major morbidity. Because a threshold of 500 U/L demonstrated appropriate discriminatory performance and was consistent with a previous report,[19] this cutoff was adopted for subsequent analyses (Supplementary Fig. S1).

We first evaluated risk factors associated with major morbidity using univariable analyses. Variables with p < 0.10 in the univariable analyses, together with clinically relevant factors previously associated with postoperative morbidity, were entered into a multivariable logistic regression model after assessment for multicollinearity. A prespecified subgroup analysis was then performed among patients with D3DA <500 U/L using the same multivariable model, excluding D3DA ≥500 U/L.

Time-to-event outcomes were evaluated using the Kaplan–Meier method and compared using the log-rank test. For these analyses, patients were stratified into four groups according to D3DA and drain culture status: D3DA low/Culture−, D3DA low/Culture+, D3DA high/Culture−, and D3DA high/Culture+.

We also conducted an exploratory analysis among patients with Culture + to evaluate postoperative outcomes according to drain management strategy (early drain removal versus prolonged drain placement).

All statistical tests were two-sided, and p < 0.05 was considered statistically significant. Analyses were performed using JMP Pro version 19.0 (SAS Institute Inc., Cary, NC, USA).

3. Results

3.1. Patients

During the study period, 460 patients underwent PD at our institution. After excluding 11 patients without drain culture data within POD3, the final analysis included 449 patients.

Baseline patient characteristics are summarized in Table 1. The median patient age was 70 years (IQR, 63–76 years), and 244 patients (54.3%) were male. The median body mass index (BMI) was 21.5 kg/m^2 (IQR, 19.5–23.9 kg/m^2). Pancreatic ductal adenocarcinoma was the most common pathological diagnosis, followed by bile duct cancer, intraductal papillary mucinous neoplasm, and duodenal papillary cancer.

Table 1. Baseline characteristics of all included patients.

Variables	n = 449
Age, years (IQR)	70 (63-76)
Sex, male n (%)	244 (54.3%)
BMI, kg/m2 (IQR)	21.5 (19.5-23.9)
Histopathology, n (%)	
Pancreatic cancer	231 (51.4%)
IPMN	45 (10.0%)
IPMC	21 (4.7%)
Bile duct cancer	51 (11.4%)
Duodenal papillary cancer	35 (7.8%)
P-NET	29 (6.5%)
Duodenal cancer	10 (2.2%)
SPN	7 (1.6%)
Chronic Pancreatitis	4 (0.9%)
Others	16 (3.6%)
Neoadjuvant therapy, n (%)	
NACRT	95 (21.1%)
NAC	81 (18.0%)
History of DM treatment	118 (26.3%)
White blood cell count, ×10^3/μL (IQR)	5.3 (4.3-6.2)
Serum albumin value, mg/dL(IQR)	3.8 (3.5-4.2)
Serum CRP value, mg/dL(IQR)	0.1 (0.1-0.2)

Abbreviations: BMI, body mass index; CRP, c-reactive protein; DM, Diabetes mellitus; IPMC, intraductal papillary mucinous carcinoma; IPMN, intraductal papillary mucinous neoplasm; IQR, interquartile range; P-NET, pancreatic neuroendocrine tumor; SPN, solid pseudopapillary neoplasm; NAC, neoadjuvant chemotherapy; NACRT, neoadjuvant chemoradiotherapy.

3.2. Perioperative outcomes according to early drain culture status

Clinicopathological characteristics and postoperative outcomes according to early drain culture status are summarized in Table 2. Early drain culture positivity (Culture+) was detected in 62 patients (13.8%) within POD3.

Table 2. Perioperative outcome.

		Culture+ (n = 62)	Culture- (n = 387)	p value
Operation time	min	519 (420.75-614.75)	507 (449-583)	0.853
Intraoperative blood loss	ml	560 (297.5-1208.75)	450 (210-841)	0.078
Blood transfusion	(%)	12 (19.4%)	54 (14.0%)	0.265
Pancreas parenchymal texture: Soft	(%)	37 (59.7%)	167 (43.3%)	0.016†
Pancreatic duct diameter	mm	3 (2-5)	4 (2-5)	0.020†
History of DM treatment	(%)	20 (32.3%)	98 (25.4%)	0.254
Preoperative biliary drainage	(%)	41 (66.1%)	158 (40.8%)	0.0002†
MIPD	(%)	13 (21.0%)	76 (19.6%)	0.807
Incomplete external pancreatic fistula	(%)	17 (27.4%)	106 (27.4%)	0.996
D1SA	U/L	286.5 (136-551.75)	250 (116-534)	0.424
D3SA	U/L	46 (22.75-117.5)	53 (27-130)	0.469
D1DA	U/L	993 (170.5-3918.5)	590 (99-2773)	0.099
D3DA	U/L	287 (96.5-1121)	167.5 (45.75-645)	0.057
Drain duration	days	4 (3-6)	4 (3-5)	0.044†
Clavien-Dindo ≥ IIIa	(%)	21 (33.9%)	57 (14.7%)	0.0002†
Post operative pancreatic fistula ≥ B	(%)	11 (17.7%)	25 (6.5%)	0.002†
Delayed Gastric Empting ≥ B	(%)	11 (17.7%)	54 (14.0%)	0.431
Thrombosis	(%)	5 (8.1%)	26 (6.7%)	0.698
Abdominal abscess ≥ IIIa	(%)	15 (24.2%)	26 (6.7%)	<0.0001†
Bile leakage	(%)	4 (6.5%)	5 (1.3%)	0.007†
Post Pancreatectomy Hemorrhage ≥ B	(%)	5 (8.1%)	8 (2.1%)	0.009†
Hospital stays	day	22 (17.75-31.5)	20 (15-28)	0.064

Abbreviations: Culture+, positive drain culture; D1SA, postoperative day 1 serum amylase; D3SA, postoperative day 3 serum amylase; D1DA, postoperative day 1 drain amylase; D3DA, postoperative day 3 drain amylase; DM, Diabetes Mellitus; MIPD, minimally invasive pancreatoduodenectomy; POD, postoperative day; S-Amy, Serous Amylase†Indicates statistical significance (p < 0.05).

Regarding operative and pancreatic factors, compared with patients with Culture–, patients with Culture + more frequently had a soft pancreatic texture (59.7% vs. 43.3%, p = 0.016) and a smaller main pancreatic duct diameter (median 3 mm vs. 4 mm, p = 0.020). History of preoperative biliary drainage was significantly more common in the Culture + group (66.1% vs. 40.8%, p = 0.0002).

Patients with Culture + also experienced significantly worse postoperative outcomes. Major morbidity occurred more than twice as frequently in the Culture + group as in the Culture− group (33.9% vs. 14.7%, p = 0.0002). Similarly, Culture+ was associated with significantly higher rates of clinically relevant POPF (17.7% vs. 6.5%, p = 0.002), intra-abdominal abscess (24.2% vs. 6.7%, p < 0.0001), bile leakage (6.5% vs. 1.3%, p = 0.007), and PPH (8.1% vs. 2.1%, p = 0.009).

3.3. Microbiological findings of early drain cultures

The microbiological profile of early postoperative drain cultures in patients with major morbidity is summarized in Fig. 2. Among the 21 patients with CD grade IIIa or higher complications, Enterococcus spp. was the most frequently isolated microorganism, followed by Enterobacter spp. and Pseudomonas spp. Polymicrobial growth was identified in two patients (Supplementary Table S1) (Fig. 2a).

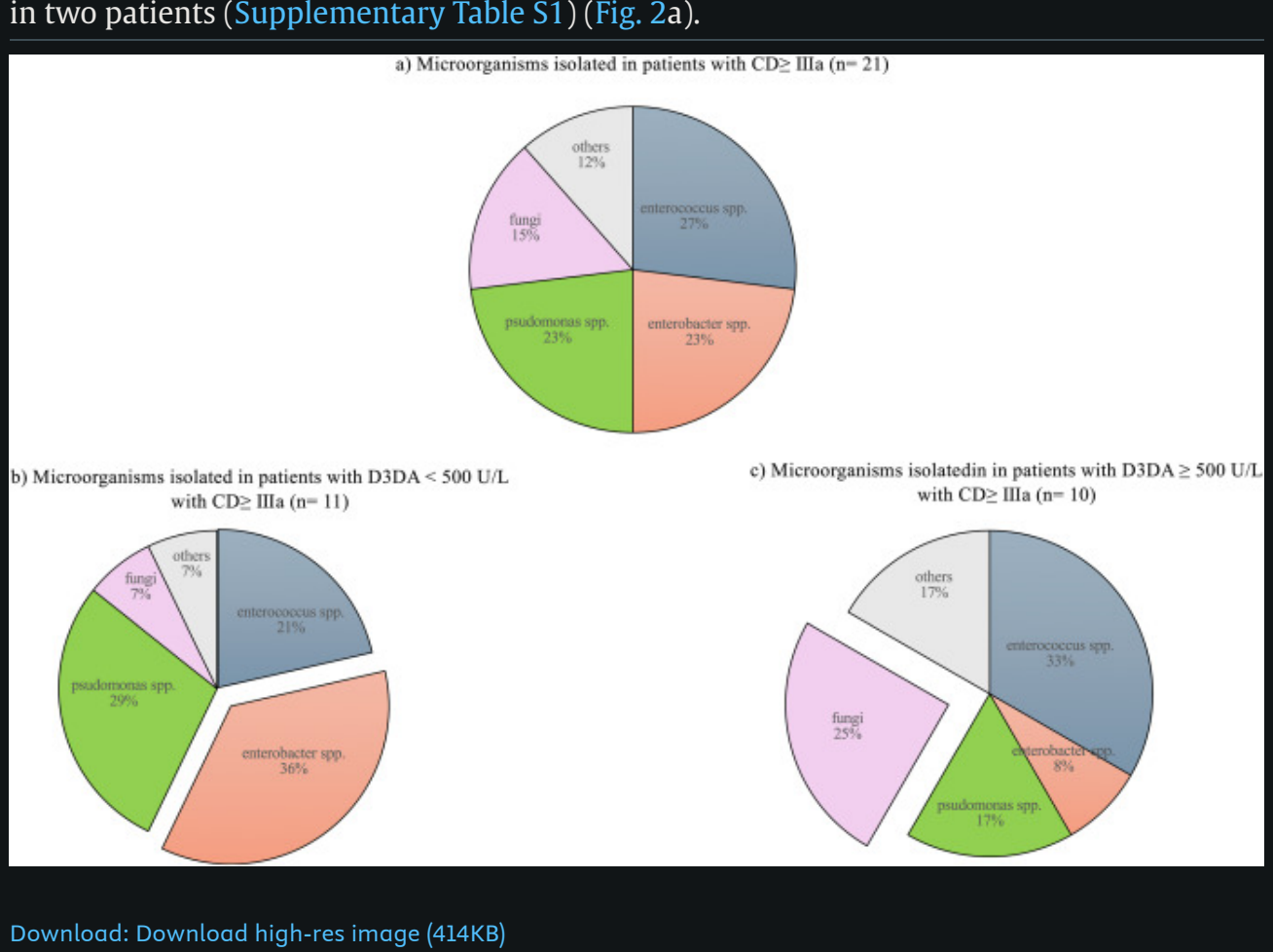

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Fig. 2. **Microbiological profiles among patients who developed major morbidity.**
Pie charts showing the distribution of microorganisms isolated from early postoperative drain cultures in patients who developed major morbidity (Clavien–Dindo grade IIIa or higher).

(A) Microorganisms isolated from all patients who developed major morbidity (n = 21).

(B) Microorganisms isolated from patients with major morbidity and POD3 drain amylase (D3DA) < 500 U/L (n = 11).

(C) Microorganisms isolated from patients with major morbidity and D3DA ≥500 U/L (n = 10).

Mixed infections were counted for each isolated microorganism.

Microbial composition differed among patients stratified according to D3DA levels of POD3. Among patients with D3DA <500 U/L (n = 11), Enterobacter spp. accounted for 36% of isolated microorganisms, compared with 23% in the overall cohort of patients with major complications (Fig. 2b). In contrast, among patients with D3DA ≥500 U/L (n = 10), fungi accounted for 25% of isolates, compared with 15% in the overall cohort (Fig. 2c). Similar trends were observed among all patients with Culture+ (Supplementary Fig. S2).

3.4. Perioperative factors associated with major morbidity

The results of the univariate and multivariable analyses for CD grade IIIa or higher postoperative complications are summarized in Table 3. In the univariate analysis, BMI ≥25 kg/m^2, incomplete external pancreatic drainage, intraoperative blood loss ≥800 mL, D3DA ≥500 U/L, and Culture+ were significantly associated with major morbidity. In multivariable logistic regression analysis, BMI ≥25 kg/m^2 (odds ratio [OR], 2.44; 95% confidence interval [CI], 1.17–5.66; p = 0.016), D3DA ≥500 U/L (OR, 2.27; 95% CI, 1.31–3.92; p = 0.004), and Culture+ (OR, 3.04; 95% CI, 1.59–5.74; p = 0.0009) remained independently associated with major morbidity.

Table 3. Univariate and multivariable analyses of risk factors for CD ≥IIIa.

Variable	Univariate analysis		Multivariate analysis	
	OR (95% CI)	p value	OR (95% CI)	p value
Previously abdominal Surgery	1.05 (0.64-1.72)	0.853	-	-
Age > 60	2.01 (0.96-4.20)	0.060	2.44 (1.17-5.66)	0.016†
BMI ≥ 25	2.30 (1.29-4.12)	0.004†	2.00 (1.06-3.71)	0.033†
PNI<41	0.82 (0.45-1.49)	0.510	-	-
mGPS ≥ 1	1.06 (0.61-1.83)	0.848	-	-
Preoperative therapy	0.69 (0.41-1.15)	0.155	-	-
MIPD	0.96 (0.52-1.77)	0.886	-	-
Incomplete external pancreatic drainage	1.74 (1.04-2.92)	0.033†	1.64 (0.93-2.85)	0.089
Portal vein resection	0.76 (0.43-1.35)	0.347	-	-
Soft pancreas	1.50 (0.92-2.45)	0.105	-	-
Bleeding > 800(ml)	2.08 (1.25-3.46)	0.004†	1.93 (1.10-3.36)	0.023†
Ope time > 600(minutes)	1.43 (0.81-2.53)	0.215	-	-
Culture+	2.97 (1.63-5.38)	0.0002†	3.04 (1.59-5.74)	0.0009†
D3DA ≥ 500 U/L	2.23 (1.35-3.70)	0.002†	2.27 (1.31-3.92)	0.004†

Abbreviations: BMI, body mass index; CD, Clavien–Dindo classification; Culture+, positive drain culture; D3DA, postoperative day 3 drain amylase; PNI, prognostic nutritional index; mGPS, modified Glasgow Prognostic Score; MIPD, minimally invasive pancreaticoduodenectomy†Indicates statistical significance (p < 0.05).

Table 4 summarizes the univariate and multivariable analyses for major morbidity among patients with D3DA <500 U/L. In the multivariable analysis, Culture+ (OR, 3.32; 95% CI, 1.39–7.62; p = 0.008) remained independently associated with major morbidity even after excluding D3DA ≥500 U/L.

Table 4. Univariate and Multivariable Analyses of Risk Factors for CD ≥IIIa in patients with D3DA < 500 U/L.

Variable	Univariate analysis		Multivariate analysis	
	OR (95% CI)	p value	OR (95% CI)	p value
Previously abdominal Surgery	1.01 (0.53-1.66)	0.833	-	-
Age >60	3.47 (1.04-11.60)	0.033†	4.91 (1.59-21.90)	0.004†
BMI ≥25	2.08 (0.95-4.59)	0.064	2.01 (0.81-4.66)	0.128
PNI <41	1.22 (0.62-2.39)	0.562	-	-
mGPS ≥1	1.43 (0.74-2.75)	0.281	-	-
Preoperative therapy	0.89 (0.47-1.67)	0.711	-	-
MIPD	0.31 (0.07-1.33)	0.095	-	-
Incomplete external pancreatic drainage	1.84 (0.96-3.53)	0.064	1.95 (0.95-3.91)	0.067
Portal vein resection	1.39 (0.73-2.64)	0.308	-	-
Soft pancreas	0.85 (0.42-1.73)	0.659	-	-
Bleeding >800(ml)	2.25 (1.18-4.27)	0.012†	1.93 (0.97-3.80)	0.061
Ope time >600(minutes)	1.86 (0.93-3.74)	0.076	-	-
Culture+	2.94 (1.34-6.46)	0.005†	3.32 (1.39-7.62)	0.008†

Abbreviations: BMI, body mass index; CD, Clavien–Dindo classification; Culture+, positive drain culture; D3DA, postoperative day 3 drain amylase; PNI, prognostic nutritional index; mGPS, modified Glasgow Prognostic Score; MIPD, minimally invasive pancreaticoduodenectomy; POPF, postoperative pancreatic fistula.†Indicates statistical significance (p < 0.05).

Corresponding sensitivity, specificity, PPV, NPV, and accuracy values for each D3DA stratum are provided in Supplementary Table S2; specificity and NPV remained high in both strata (≥86% and ≥77%, respectively), while sensitivity was modest, reflecting the multifactorial etiology of major morbidity after PD.

3.5. Time-to-event analysis

The Kaplan–Meier curves for time to CD grade IIIa or higher postoperative complications according to early drain culture status are shown in Fig. 3a. Patients with Culture + demonstrated a significantly higher cumulative incidence of major morbidity than those with Culture– (33.9% vs 14.7%, log-rank p = 0.0002).

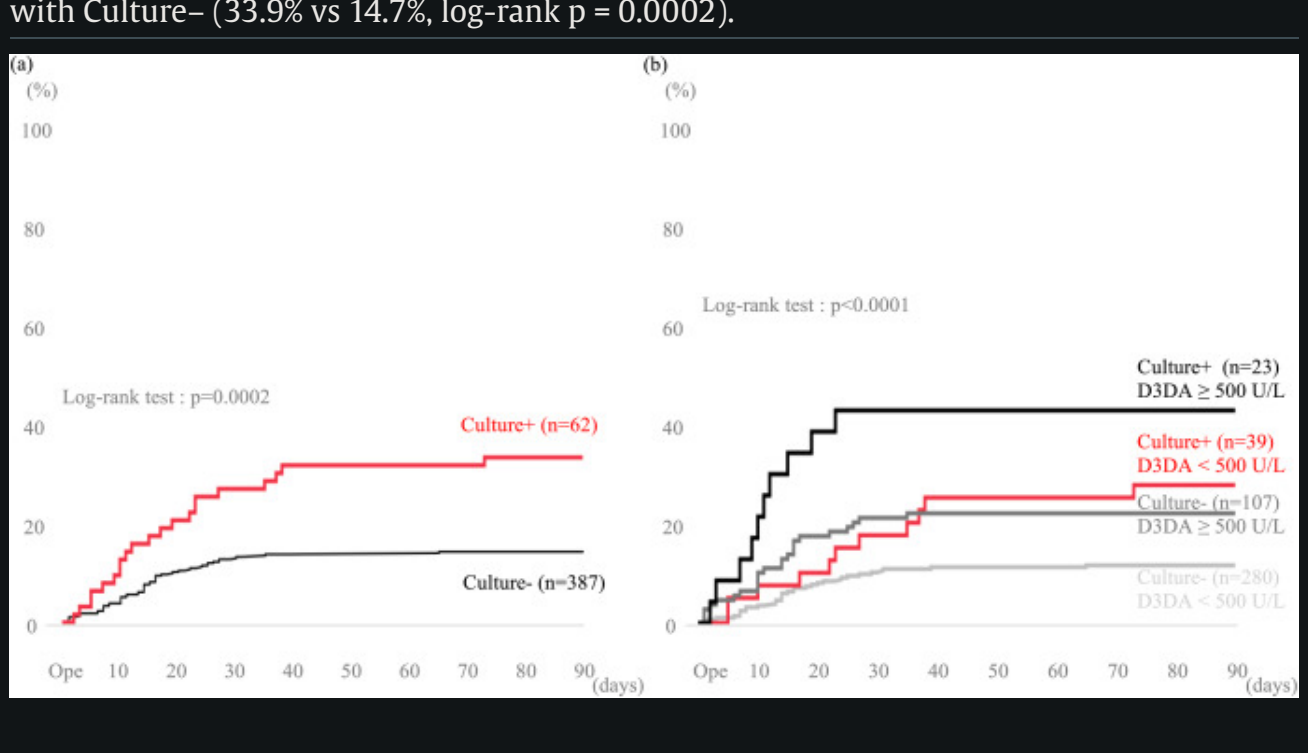

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Fig. 3. **Time-to-event analysis according to POD3 drain amylase levels and drain culture status.**
(a) Kaplan–Meier curves showing the cumulative incidence of Clavien–Dindo grade IIIa or higher postoperative complications stratified by drain culture status. Log-rank testing was used for group comparisons.

(b)Kaplan–Meier curves showing the cumulative incidence of Clavien–Dindo grade IIIa or higher postoperative complications stratified by POD3 drain amylase (D3DA) levels and drain culture status. Log-rank testing was used for group comparisons.

To evaluate the combined prognostic value of D3DA levels and drain culture status, patients were stratified into four groups (Fig. 3b). The cumulative incidence of major morbidity was 43.5% in the D3DA ≥500 U/L/Culture + group, 22.4% in the D3DA ≥500 U/L/Culture− group, 28.2% in the D3DA <500 U/L/Culture + group, and 11.8% in the D3DA <500 U/L/Culture− group (log-rank p < 0.0001). Among patients with low D3DA levels, Culture+ was associated with a later onset of postoperative complications (median POD 23, IQR 10–37 days) and included several major complications (Supplementary Table S3, Supplementary Fig. S3).

3.6. Exploratory analysis of drain management in patients with Culture+

Among the 62 patients with Culture+, 38 (61.3%) underwent early drain removal (≤POD4), whereas 24 (38.7%) required prolonged drain placement (≥POD5). Major morbidity occurred in eight patients (21.1%) in the early drain removal group and 13 (54.2%) in the prolonged drainage group (Supplementary Table S4). Clinically relevant POPF, intra-abdominal abscess, bile leakage, and PPH were also more frequently observed among patients requiring prolonged drain placement. Because drain removal was determined according to the postoperative clinical course, these findings should be considered descriptive and hypothesis-generating rather than indicative of a causal effect of drain management strategy.

4. Discussion

This study evaluated the clinical significance of early postoperative drain culture positivity in one of the largest contemporary single-center cohorts undergoing PD at a high-volume pancreatic surgery center in Japan. We specifically investigated whether drain culture positivity within POD 3 was associated with major morbidity. Our findings demonstrated that early drain culture positivity is independently associated with major morbidity after PD. Importantly, this association persisted among patients with low D3DA levels, suggesting that postoperative infectious risk may not be fully explained by pancreatic leakage alone. Thus, early surveillance drain cultures may provide complementary prognostic information beyond conventional drain amylase-based risk stratification. However, drain culture positivity itself does not necessarily indicate a clinically relevant infection. In the present study, we consider drain culture positivity as a potential early marker of an unfavorable postoperative course and a complementary marker to conventional risk assessment.

Postoperative management after PD has traditionally centered on detecting CR-POPF, with D-AMY as the principal biomarker used for postoperative risk assessment and early drain removal.[9], [10], [20], [21] Although D-AMY remains an established predictor of pancreatic leakage, it primarily reflects biochemical evidence of pancreatic fistula and may not adequately identify patients who subsequently develop infectious complications through other mechanisms. In the present study, elevated D-AMY was primarily associated with fistula-related complications, whereas Culture+ was associated with a broader spectrum of major morbidity. Even in patients with low D-AMY levels, those who were Culture + still developed late postoperative events, suggesting the presence of infection-related risk that cannot be captured by biochemical leakage markers alone. Consistent with this, the specificity and NPV of Culture + for predicting major morbidity remained high even in the low-D3DA subgroup, supporting its potential utility as a complementary risk marker in patients not identified as high-risk by D-AMY alone.

The microbiological findings support this hypothesis. *Enterococcus* spp., *Enterobacter* spp., and *Pseudomonas* spp. were the predominant organisms isolated from early postoperative drain cultures. *Enterococcus* and *Enterobacter* species are common constituents of the biliary and intestinal microbiota, consistent with early contamination of the operative field following biliary and gastrointestinal reconstruction.[22], [23], [24] In contrast, *Pseudomonas* spp. are well-known healthcare-associated pathogens capable of biofilm formation, potentially reflecting persistent drain colonization.[25], [26], [27] The microbiological profile observed in this study may also reflect the selective pressure of perioperative FMOX prophylaxis, which has limited activity against *Enterococcus* spp. and *Pseudomonas* spp. Additionally, fungal organisms, including *Candida* spp. were more frequently identified in

Additionally, fungal organisms, including *Candida* spp, were more frequently identified in patients with high D3DA, consistent with the report by Sato et al., which linked fungal contamination to CR-POPF.[28] Conversely, *Enterobacter* spp. accounted for a greater proportion of isolates among patients with low D3DA levels. As representative enteric microorganisms, these findings suggest that persistent bacterial colonization of the postoperative abdominal cavity may contribute to sustained inflammation and delayed infectious complications, including intra-abdominal abscess and PPH, even in patients without elevated D3DA levels. A positive drain culture may reflect bacterial colonization rather than clinically significant infection, and microbiological findings alone cannot distinguish between these states. However, the association between Culture+ and major morbidity, independent of D-AMY, suggests that at least a subset of positive cultures reflect a clinically relevant bacterial burden rather than incidental colonization.

Importantly, drain culture positivity remained associated with major morbidity among patients with low D3DA levels. This finding suggests that early drain culture positivity may provide prognostic information beyond what is captured by drain amylase levels alone, at least in this subgroup. The co-occurrence of Culture+ with delayed-onset complications and the microbiological findings described above raises the possibility that persistent intra-abdominal bacterial colonization, whether biliary- or gut-derived, or related to healthcare-associated pathogens, may contribute to postoperative morbidity independent of amylase-detectable pancreatic leakage.29, 30, 31.

These findings have several potential clinical implications. Current postoperative management strategies primarily rely on D-AMY to guide early drain removal.8, 9 Although exploratory in nature, our findings suggest that surveillance culture positivity alone should not prompt deviation from current early drain removal strategies, particularly given that drain removal was individualized based on the patient's clinical condition and drain findings, with patient safety and adequate source control prioritized. Instead, our results suggest that D-AMY alone may be insufficient to assess infectious risk and that patients with Culture + warrant closer postoperative surveillance. Such surveillance could include serial inflammatory marker assessment, a lower threshold for imaging evaluation, and intensified outpatient follow-up to facilitate earlier recognition of delayed infectious complications.32, 33 This supports the potential value of a culture-guided surveillance strategy after PD. However, positive surveillance cultures should not necessarily prompt intervention in clinically well patients. Further studies are needed to clarify whether culture-guided management improves outcomes.

This study has several limitations. First, the retrospective single-center design introduces potential selection and management biases. Second, although drain culture sampling followed institutional protocols, postoperative management was not fully standardized. Furthermore, postoperative drain culture positivity may be influenced by factors such as surgical technique and drain type or management. Third, Culture+ was defined by the isolation of any microorganism, without distinguishing infection from asymptomatic colonization, as culture alone cannot reliably differentiate the two. This potential misclassification may have attenuated the observed association. Further studies incorporating quantitative culture or inflammatory biomarkers are warranted. Finally, analyses of postoperative antimicrobial use and drain management were subject to indication bias because these interventions were determined by the evolving clinical course and may themselves satisfy the criteria for CD IIIa complications. Therefore, the clinical impact of culture-guided interventions remains unclear and warrants prospective investigation.

5. Conclusion

Our findings suggest that routine early drain culture assessment may provide clinically actionable information beyond conventional biochemical markers. Patients with Culture+, particularly those without elevated drain amylase levels, may benefit from closer postoperative monitoring and individualized management to detect complications before clinical deterioration occurs. Early drain culture assessment may complement conventional drain amylase evaluation by identifying a subset of patients at risk for infectious morbidity despite the absence of biochemical evidence of pancreatic leakage.

Ethics declaration

Verbal informed consent to take part in the study and to publish the article was obtained from all participants or their legal representatives. The study was approved by the Ethics Committee of the Graduate School and Faculty of Medicine, Kyoto University (approval no.: R1721) and was carried out in compliance with the Helsinki Declaration. An opt-out approach was adopted. The privacy rights of participants have been observed.

This study was performed in compliance with relevant laws, regulatory frameworks and guidelines where the research took place. This study was approved by the Kyoto University. (Approval No. R1721)

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Author contributions

K.K. wrote the manuscript and tables. K.Y., K.N.,and E.H. supervised the whole editing process. K.K., K.Y., K.N., S.I., K.N., N.M., S.N., A.S., S.O., Y.U., T.I., T.I., and E.H. were involved in the conception, design, and conduct of the study, and analysis and interpretation of the results. K.K., K.Y., and S.I. were involved in data collection. K.K., and K.Y. were responsible for the statistical analyses. K.K.and K.Y. wrote the first draft of this manuscript. E.H. supervised this project. All authors have edited, reviewed, and approved the final version of the manuscript.

Declaration of competing interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:Etsuro Hatano serves as a section editor in Hepato-biliary-pancreatic Surgery for the Asian Journal of Surgery. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following is the Supplementary data to this article:

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Multimedia component 1.

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